Bank Customer Churn Analysis

Analytics in Practice Workshops

MSc Business Analytics | Trinity College Dublin

LOST REVENUE AND HIGH CUSTOMER ACQUISITION COSTS DUE TO CHURN

Identify key drivers of churn and predict high-risk customers for bank, leveraging:

- Customer demographics
- Financial data
- Engagement behaviour

Primary objective: Reduce churn rate by 15% through targeted, data-driven retention strategies

Link to GitHub REPO

The bank is **losing customers** (ca.20% churn rate), which directly **impacts revenue** and **increases acquisition costs**.

Importantly, **churners** are not low-value customers, but are financially stable, high-balance **individuals leaving by choice**, indicating a **service and engagement problem** rather than a credit risk issue.

DATA OVERVIEW

DATASET SIZE:
10,000+ CUSTOMER RECORDS

Target Variable:
churn (1 = customer churned, 0 = stayed)

KEY FEATURES:

- ➔ **Demographics**:
  - customer_id
  - credit_score
  - country (France, Spain, Germany)
  - gender
  - age
- ➔ **Account Info:**
  - tenure (years with bank)
  - Balance
  - products_number (bank products held)
  - credit_card (has credit card? 0/1)
  - active_member (0/1)
- ➔ **Financial:**
  - estimated_salary

- ➔ **Data Types:**
  - Mix of numerical (int/float) and categorical (country, gender, binary flags)
- ➔ **Missing Values:**
  - No obvious missing values in the sample (balance can be 0, which is valid)
- ➔ **Potential Imbalance:**
  - Churn appears less frequent than non-churn (needs verification)

****Use Case:** Binary classification to predict customer churn based on banking behaviour and demographics

Input: 10,000 raw customer records with 12 columns

DATA CLEANING & VALIDATION

Range assertions, null checks, and duplicate removal

FEATURE ENGINEERING

Created 4 new domain-meaningful features

Captures behavioral signals (e.g., balance exposure, product engagement)

TRAIN/TEST SPLIT

80/20 split using stratified sampling (preserves churn ratio)

Preprocessing with ColumnTransformer

Numeric scaling and categorical encoding applied

Handling Class Imbalance

SMOTE applied to training set only

Corrects 80/20 imbalance → achieves 50/50 balanced training set

Output: processed CSV files, balanced target labels, fitter preprocessor object

Key Drivers of Churn

Demographics (High-Value Risk Segment):
Age:
- 50–60 → highest churn (~56%)
- <30 → lowest (~7%)

Gender:
- Females churn more (~25% vs ~16%)

→ Strongest predictors
→ High-value customers are at higher risk

Engagement & Product Usage:
- Products = 2nd most important feature
- Active members → lower churn

Non-linear insight:
- 1 product → ~27% churn
- 2 products → ~8% (optimal)
- 3-4 products → 80–100% churn

→ Sweet spot = 2 products
→ Over-selling leads to disengagement & churn

Geography (Concentrated Risk):
- Germany: ~32% churn (highest)
- France & Spain: ~16% (stable)

→ Churn is market-specific, not bank-wide
→ Likely driven by pricing, competition, expectations

Lifecycle Behaviour:
- High churn at onboarding (~23%)
- Decreases mid-tenure
- Increases again at later stages (~year 9)

→ Retention must be continuous across lifecycle (not just at acquisition)

Insights & Segmentation (*Why It Matters*)

Balance Behaviour (Hidden Risk):

- High-balance customers still churn
- Churned customers have higher median balance (~€100k)
- Zero-balance customers churn less

→ Engagement > wealth
→ Bank is losing its most valuable customers

Model Performance (Predictive Power):

- Logistic Regression: AUC: 0.79 (baseline)
- Random Forest: AUC: 0.86 (best)

→ Can reliably identify high-risk customers
→ Enables targeted retention (lower cost, higher impact)

Customer Segments (Where to Act):

Cluster 0 (Under-engaged, high balance)

- Similar risk, moderate churn

Cluster 1 (New customers)

- "Free trial phase", early churn risk

Cluster 2 (Low value, stable)

- Low churn, low revenue

Cluster 3 (High value, high risk)

- High balance, low engagement → top priority

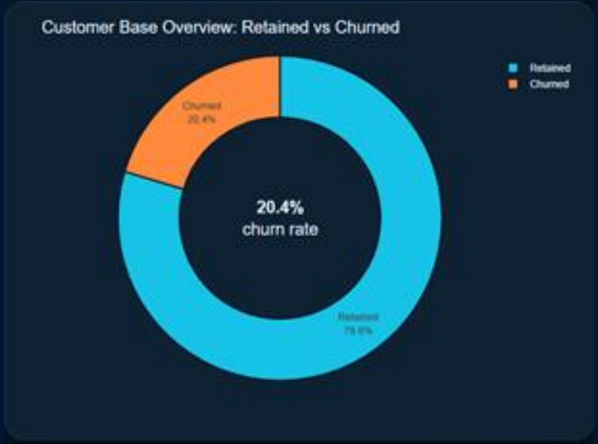
Data Relationships:

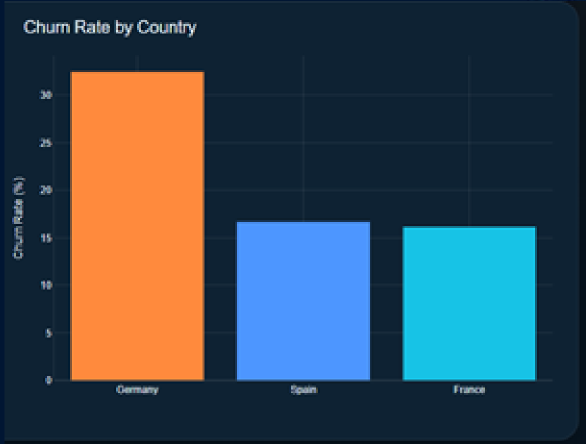
- Weak linear correlations → non-linear patterns

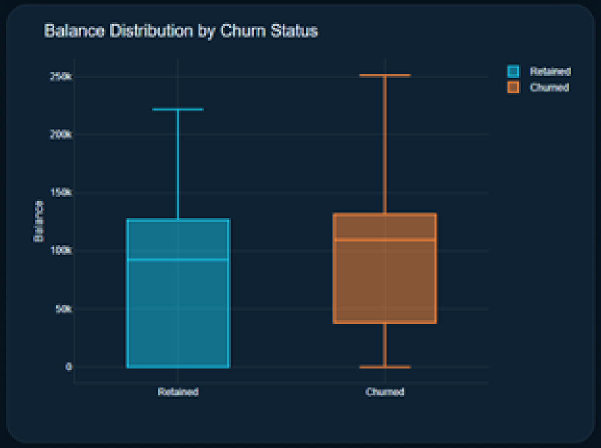
→ Explains why Random Forest performs better than Logistic Regression

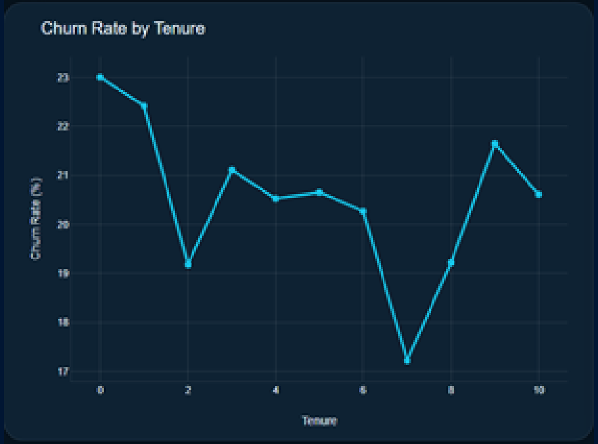
Key Insight:
→ Highest churn risk = high balance + low engagement
→ Churn is predictable, segment-driven, and actionable

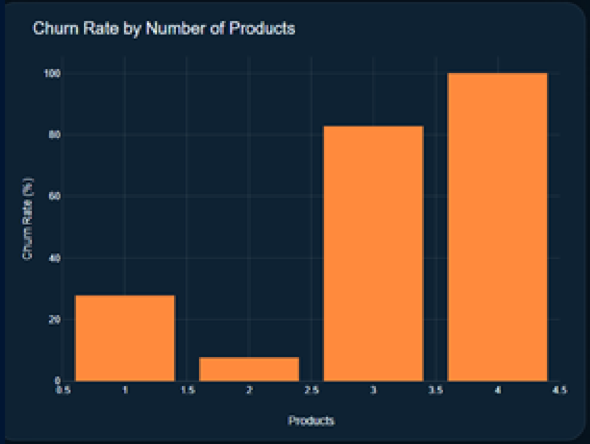
Total Customers	Churned Customers	Churn Rate	Avg Balance	Avg Credit Score	Avg Estimated Salary
10,000	**2,037**	**20.4%**	**€76,486**	**651**	**€100,090**

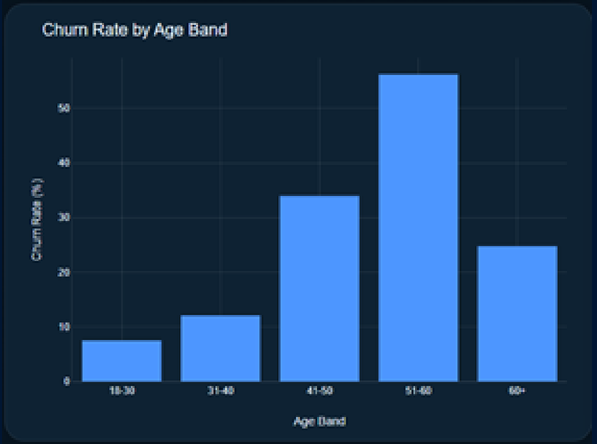












Retention Strategy I (*What Should We Do*)

1. Protect High-Value Customers:
Target Clusters 0 & 3 (high balance, high risk)
Increase stickiness through:
- Personalised offers (wealth, investment)
- Product bundling (savings + credit)
- Relationship management

2. Optimise Product Strategy:
Target 2 products per customer (optimal retention point)
Avoid over-selling (3+ products → churn risk)
Move customers from 1 → 2 products via:
- Bundles
- Incentives
- Personalised recommendations

3. Lifecycle Retention:
- Early (0–12 months): onboarding, education, engagement nudges
- Mid: personalised offers, usage-based engagement
- Late: loyalty programs, proactive outreach

4. Increase Engagement:
Activate inactive users through:
- App nudges & reminders
- Cashback / usage rewards

Focus on *active_member* → key churn driver

Retention Strategy II & Business Impact

5. Segment-Based Actions:

Cluster 1 (New customers):
- 90-day onboarding journey
- Check-ins & personalised offers

Germany (High churn market):
- Investigate pricing & competition
- Localised campaigns

Age-based strategy:
- 50+ (high risk): wealth planning, dedicated support
- Younger: digital-first, gamified engagement

Revenue Growth:
- Retaining 10–15% of high-value customers → significant revenue gain
- Prevents loss of high-balance accounts

Cost Optimisation:
- Random Forest enables targeted retention
- Reduces spend on low-risk customers

Increased Customer Lifetime Value:
- More engagement →
  - Higher retention
  - More cross-sell revenue
  - Longer customer lifespan

Strategic Shift:
- From customer acquisition
- To retention & relationship deepening

LLM Integration & Key Learnings

LLM Workflow Integration

1.Code Co-pilot:
Generated ML pipelines (SMOTE + scaling)
→ Refined to avoid data leakage using cross-validation pipelines

2.Insight Translation:
Converted technical outputs
→ Feature importance, model metrics → business narratives

3.Strategy Support:
Helped translate clustering results
→ Into targeted retention campaigns

Human Accuracy Checks

1.Data Leakage Fix:
Corrected flawed preprocessing
→ Ensured valid model performance (AUC 0.855)

2.Anti-Hallucination:
Verified formulas & outputs against dataset

3.Source Validation:
Checked AI references
→ Identified & corrected hallucinated sources

Prompt Engineering

Low-context prompts → generic outputs

Improved by:

- Adding specific metrics (Recall 0.65, Precision 0.57)
- Defining role ("Senior Analytics Consultant")
- Providing dataset context (clusters, features)

→ Result: More accurate, business-ready insights

Key Lessons Learned

1.Context = Quality:
More detail → better outputs

2.AI ≠ Autonomous:
Requires validation & correction

3.Audit > Prompting:
Human role is to challenge & verify AI outputs

4.Biggest Value:
Bridging technical analysis → business decisions

Processes & Reflections

Key Takeaways:

- Customer churn is primarily driven by **low engagement** and **suboptimal product usage** (especially among specific age groups, lifecycle stages, and regions), **not just financial value.**
- This highlights the **need for targeted, segment-based retention strategies**, namely for high-value but under-engaged customers; even high-value customers churn if they are not deeply engaged.
- Additionally, **churn is not evenly distributed**, but rather, it **clusters** around identifiable patterns across lifecycle stage (early and late tenure), product usage, and geography, enabling precise intervention rather than broad retention efforts.

Category	Key Points
What Went Well	• Clean dataset → smooth & efficient EDA • Correct SMOTE implementation inside CV pipeline • Clear role division → strong collaboration & handoffs
Challenges Faced	• Class imbalance required careful handling • Coordinating across multiple sub-teams
What We'd Do Differently	• Use LightGBM / XGBoost (more powerful models) • More systematic hyperparameter tuning • Earlier feature engineering
Key Learnings	• Avoid data leakage (SMOTE inside pipeline, not before) • Use stratified splits for imbalanced data • Translate results into business impact (€), not just accuracy